Dóra Cziborová

Computer scientist · PhD student

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Education and Degrees

2024–present **Computer Science PhD**, *Budapest University of Technology and Economics*, Critical Systems Research Group (ftsrg), Research topic: Formal Verification of Real-time Software-based Systems.

2022–2024 **Computer Science Engineering MSc**, *Budapest University of Technology and Economics*, specialization in Critical Systems, graduated with highest honors, participant of the IMSc program.
Master's thesis: Abstraction-based Timed Model Checking for Software-intensive System Models

2018–2022 **Computer Science Engineering BSc**, *Budapest University of Technology and Economics*, graduated with highest honors, participant of the IMSc program.
Bachelor's thesis: Generalizing Lazy Abstraction Refinement Algorithms with Partial Orders

Skills

Research	Formal Methods, Model Checking, Timed Systems, Lazy Abstraction, CEGAR
Development	Java, Kotlin, Python, Prolog, Elixir, C#, R, SQL, C, C++, Git
Languages	Hungarian (native), English (advanced), Slovak (advanced), Czech (passive), German (passive)

Publications

2026 **Unified timing-aware program verification**, Dóra Cziborová, Mihály Dobos-Kovács, Kristóf Marussy, András Vörös, *Fundamental Approaches to Software Engineering (FASE)*.

2026 **Sequential and parallel implementation of lazy abstraction strategies**, Mihály Dobos-Kovács, Dóra Cziborová, András Vörös, *Acta Cybernetica*.

2025 **Automata-based representation of coordination for distributed reactive systems**, Richárd Szabó, Dóra Cziborová, *International Conference on Formal Methods and Foundations of Artificial Intelligence (FMF-AI)*.

2025 **Towards configurable coordination for distributed reactive systems**, Richárd Szabó, Dóra Cziborová, *32nd Minisymposium of the Department of Measurement and Information Systems of the Budapest University of Technology and Economics*.

2024 **Modeling of time-dependent behavior in fault-tolerant systems**, Dóra Cziborová, Richárd Szabó, *31st Minisymposium of the Department of Measurement and Information Systems of the Budapest University of Technology and Economics*.

Open Source Contributions

2021–present **Theta**, *a generic, modular and configurable formal verification framework supporting various formalisms and algorithms.*

Experience

2024 **16th Alpine Verification Meeting (AVM)**, *Freiburg im Breisgau, Germany*.
Presentation: Abstraction-based Model Checking for Real-time Software-intensive System Models

2024 **"Addressing Verification and Validation Challenges in Future Cyber-Physical Systems" (ADVANCE) H2020 Research and Innovation Staff Exchange (RISE) research project**, *ResilTech S.R.L., Pontedera, Italy*.

2023 **15th Alpine Verification Meeting (AVM)**, *Prague, Czech Republic*.
Presentation: Combining CEGAR and Lazy Abstraction for Verifying Timed Systems

Teaching

2024–present **Automated Verification Techniques**, *MSc course*, delivering lectures on SMT solvers.

2024–present **Software Engineering**, *BSc course*, delivering laboratory practices, scoring exams.

2023–present **Formal Methods**, *MSc course*, delivering lectures on model checking, assembling and grading homework assignments, scoring exams.

2024–2025 **System Optimisation**, *MSc course*, scoring exams.

2023–2024 **Languages and Automata**, *MSc course*, scoring exams.

2020–2021 **Databases**, *BSc course*, delivering classroom practices, scoring exams.

2019 **Basics of Programming 1**, *BSc course*, delivering laboratory practices.

Awards and Scholarships

2025 **3rd place at the National Scientific Students' Associations Conference (OTDK)**.

2025-2027 Doctoral Excellence Fellowship Programme Scholarship (DKÖP)

2023 **1st place at the Scientific Students' Associations Conference (TDK)**,
Abstraction-based model checking for real-time software-intensive system models.

2023 National Academic Scholarship (NFÖD)

2022 **2nd place at the Scientific Students' Associations Conference (TDK)**,
Abstraction-based model checking techniques for real-time systems (D. Cziborová, B. Á. Vizi).

2022 Dean's IMSc Scholarship

2020 fall, 2021 spring, 2021 fall, 2022 spring, 2023 spring Scholarship of the Faculty of BME-VIK (KBME)